Campus Booking System

Modernize Room and Facility Booking on Campus

An intuitive and elegant solution for managing campus spaces.

Introduction & Overview

The Campus Booking System is a centralized, responsive web application designed to simplify and streamline the reservation of campus facilities.

Core Objectives:

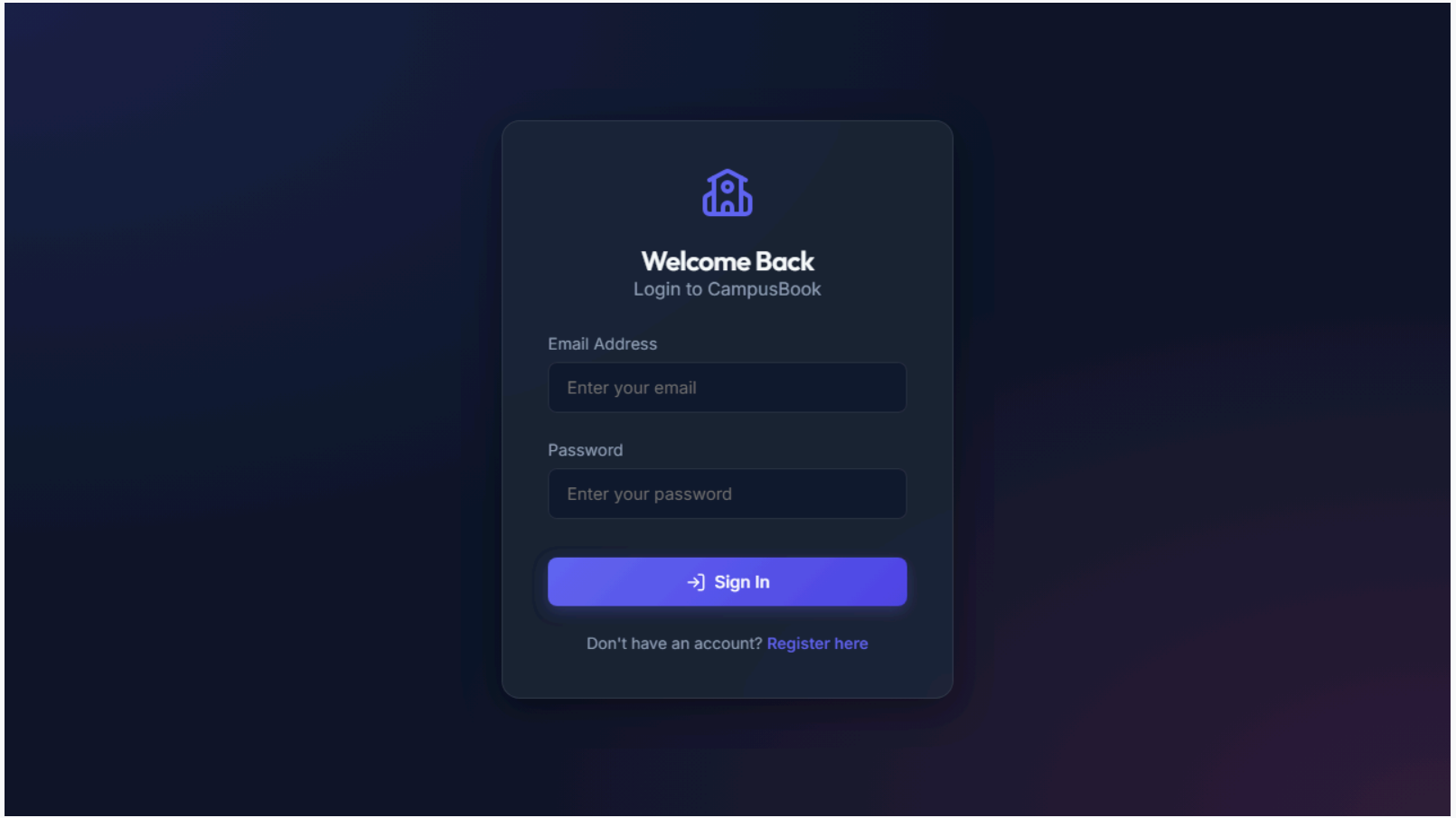
- **Eliminate** scheduling conflicts
- **Provide** real-time availability of rooms
- **Deliver** a premium, user-friendly interface with glassmorphism design elements

Features Available

- **Real-Time Booking:** Browse and book rooms dynamically.
- **Role-Based Access Control:** Distinct interfaces and privileges for Admins and Students/Staff.
- **Smart Validation:** Prevents invalid durations and respects facility operating hours.
- **Responsive Design:** Stunning UI featuring glassmorphism and micro-animations.
- **Secure Authentication:** JWT-based secure login and registration system.

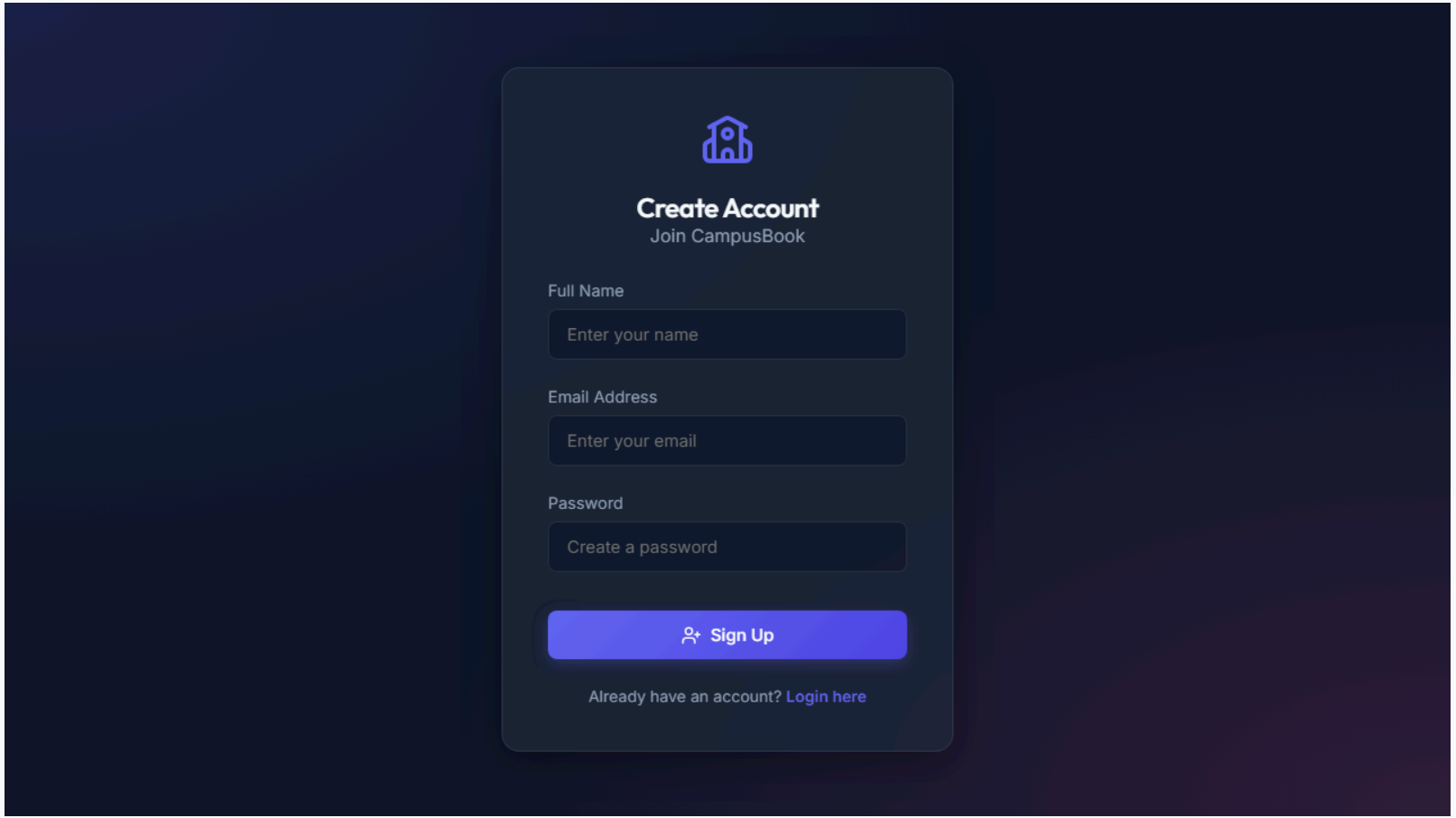
Authentication & Access (Login)

The platform provides a secure entry point for all users, distinguishing between standard users and administrators automatically.



User Registration

New users can easily sign up and create a profile. Special admin privileges are assigned based on specific criteria.



Functions of Admin

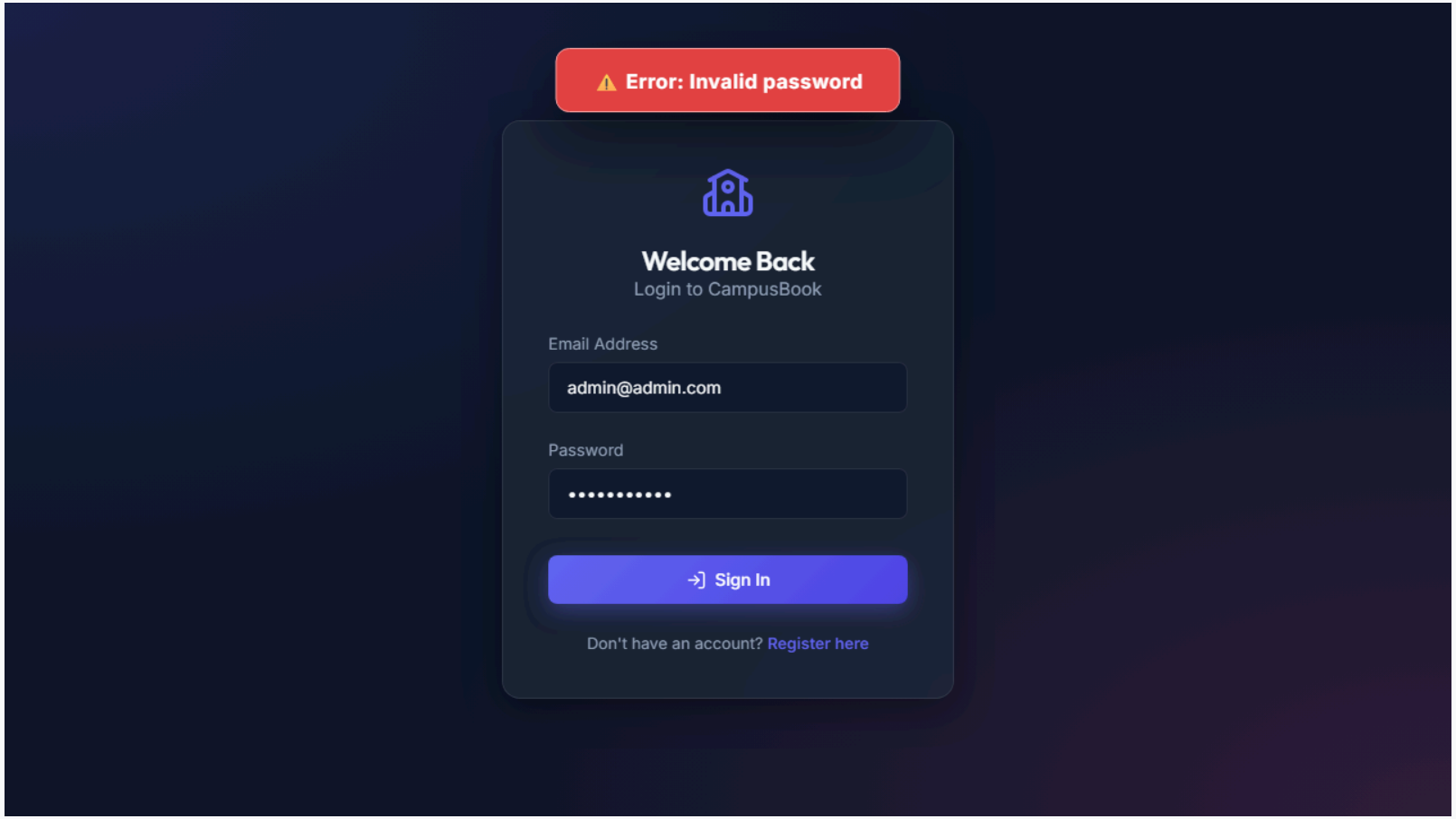
Administrators have full oversight and control over the campus facilities and booking requests.

Key Responsibilities:

- **Manage Facilities:** Add, update, or remove campus rooms and facilities.
- **Monitor Usage:** View an overview of system utilization and room occupancy.
- **Process Requests:** Approve, reject, or cancel pending booking requests.
- **Filter & Search:** Functional filtering to easily manage various booking statuses.

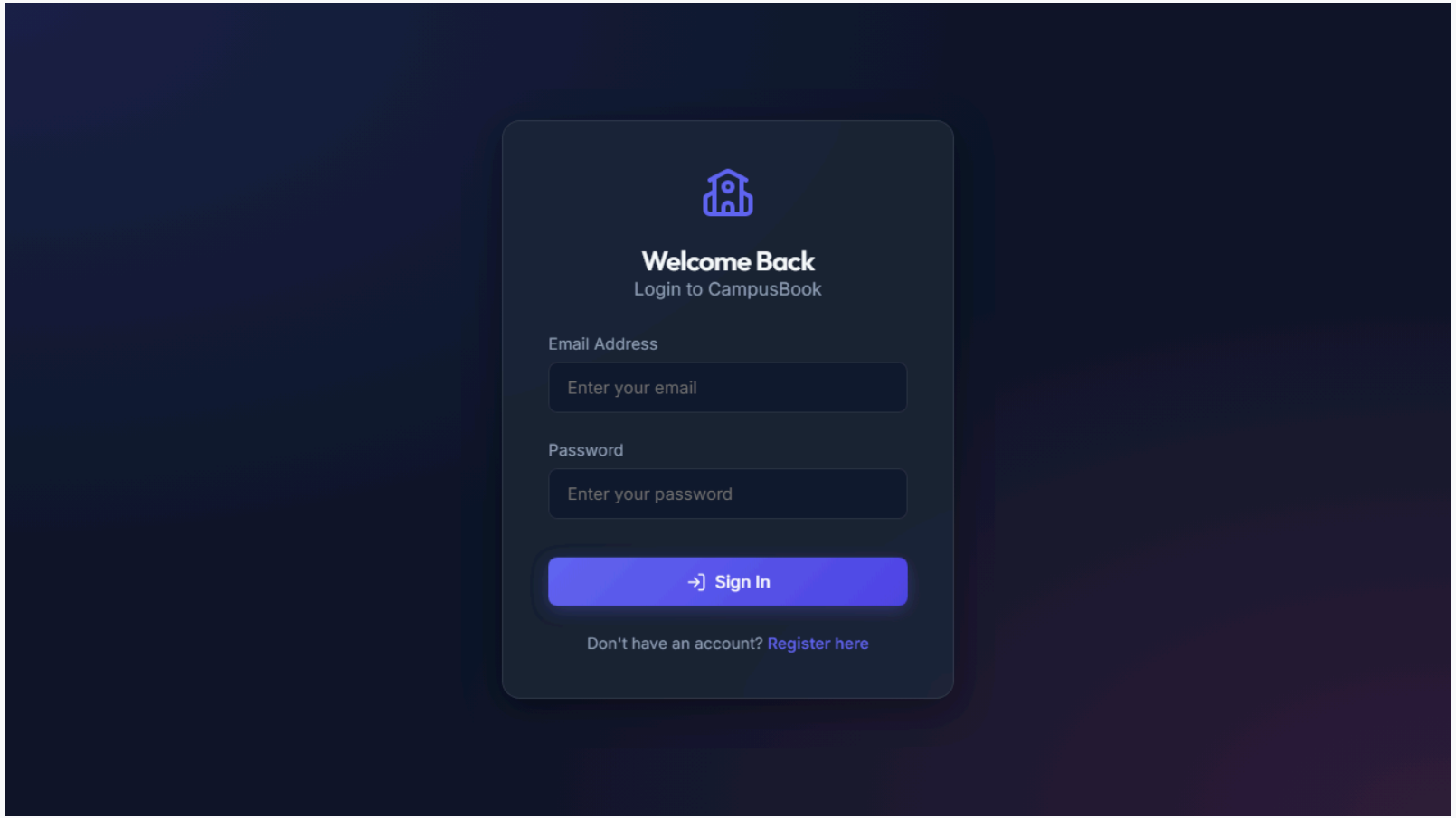
Admin Dashboard

A centralized hub for administrators to monitor pending requests and system statistics.



Admin Room Management

Admins can easily maintain the directory of available rooms, updating capacity, type, and availability.



Functions of Student & Staff

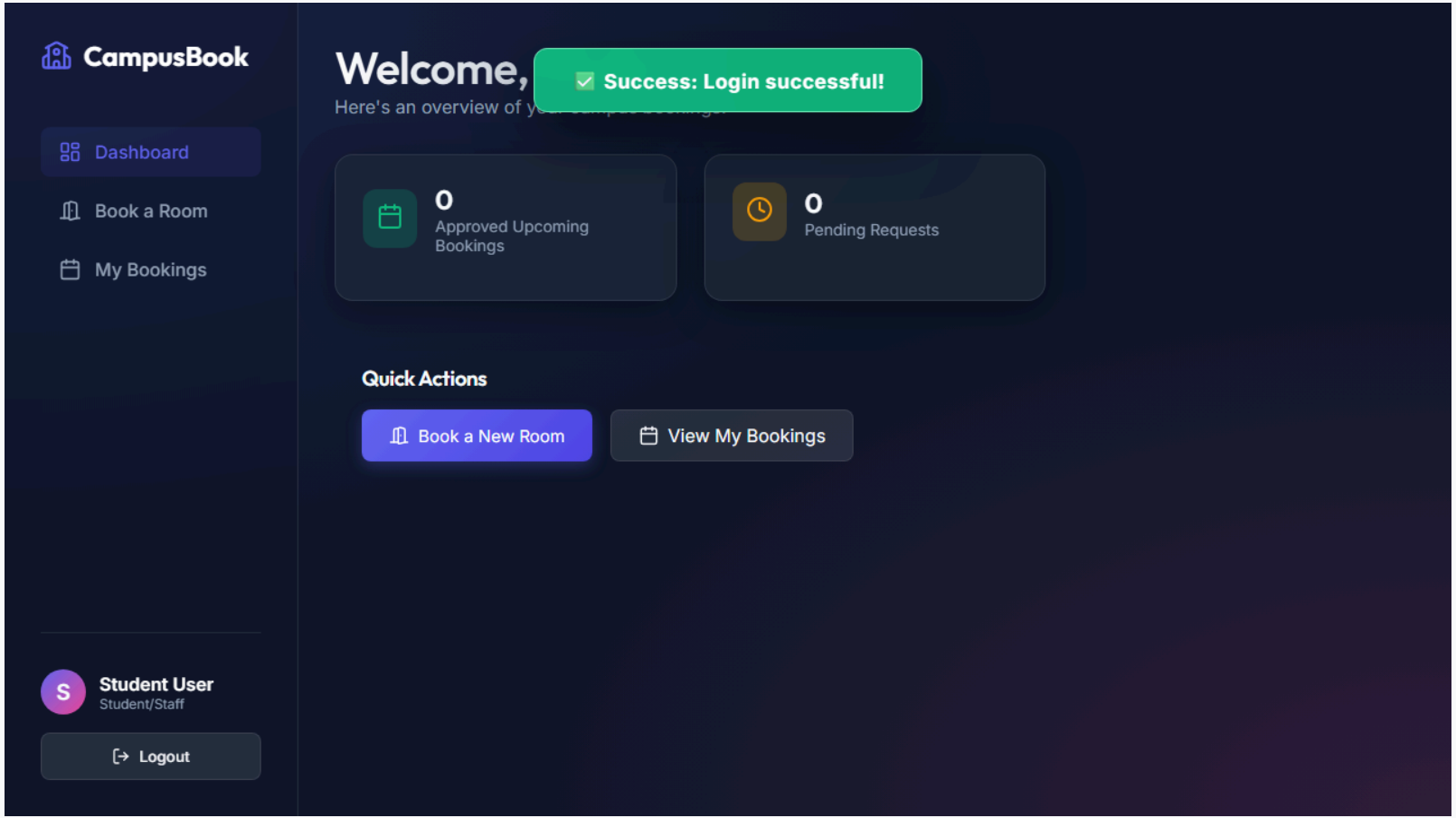
Standard users can easily find and reserve the spaces they need for their academic and extracurricular activities.

Key Capabilities:

- **Search Rooms:** Filter rooms by capacity, type, and availability.
- **Book Facilities:** Select dates and times to reserve a room effortlessly.
- **Modify Bookings:** Update or cancel existing reservations.
- **Track Status:** View the real-time approval status of past and upcoming bookings.

Student / Staff Dashboard

A personalized view showing upcoming bookings, past history, and quick access to new reservations.



System Testing

Ensuring reliability, security, and correctness across the entire platform.

Unit Testing:

- Individual components (e.g., date-time validators, authentication helpers) are tested in isolation.
- Ensures backend API validations and frontend UI elements behave correctly under edge cases.

Integration Testing:

- Tests the interaction between the React frontend, Node/Express backend, and MongoDB database.
- Verifies complete end-to-end workflows, such as the full room booking lifecycle and admin approval processes.

Conclusion

The Campus Booking System provides a seamless, efficient, and visually stunning experience for managing campus facility reservations.

Empowering users with control and administrators with insight.